

Reducing Drive Shaft Vibrations in Gulo 3000 Prototype Hybrid Electric Model

Evers Baskey, Sophia Dragich, Evan Link, Vrinda Thakur

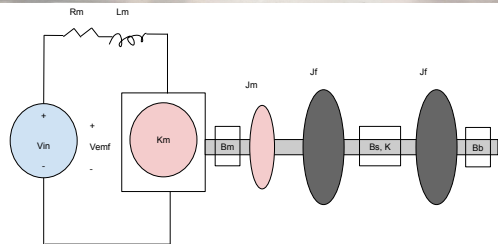
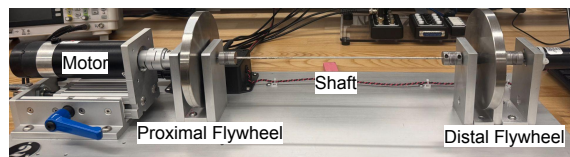


Background

Mesa Hybrid Vehicles Inc. developed a new prototype hybrid electric model, the Gulo 3000. The prototype experienced temporary vibrations when the motor was turned on and off that were so large they disrupted passengers. Mesa attributed these vibrations to the drive shaft which connects the DC motor to the front wheels of the car. Mesa requested that we determine if the vibrations were due to the shaft's flexibility, characterize the system, develop a mathematical model, and provide the four most effective solutions to reduce the vibrations by least 30%.

Methods

Experimental setup and schematic drawing



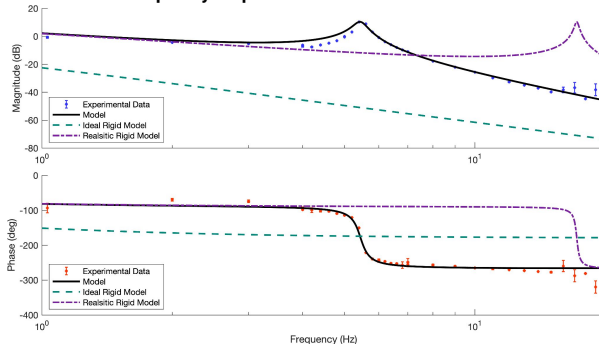
Electrical: R_m motor resistance, L_m motor inductance, V_{in} input voltage, V_{emf} motor back-EMF.
Mechanical: B_m motor damping, J_m motor inertia, J_s flywheel inertia, B_s shaft damping, K torsional spring constant, B_b bearing damping.

Experimental Characterization

The system was characterized using DC motor tests, free-response measurements, and frequency-response testing. R_m and K_m were determined experimentally, while flywheel inertia and shaft stiffness were calculated from provided dimensions and material properties. Free-response tests with each side locked were used to estimate B_m and B_b . Frequency-response measurements from 1 to 20 Hz with finer resolution near resonance were then compared with a lumped-parameter model of the system. After refining the effective stiffness and damping distribution, sensitivity analysis was used to identify promising design parameters, which were varied to evaluate their ability to reduce the resonant amplitude by at least 30%.

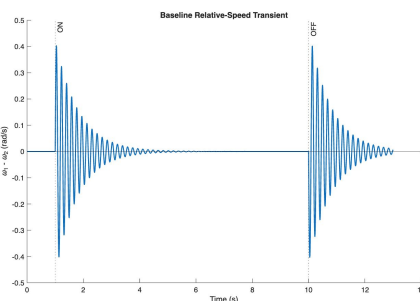
Model Validation

Flexible shaft frequency response



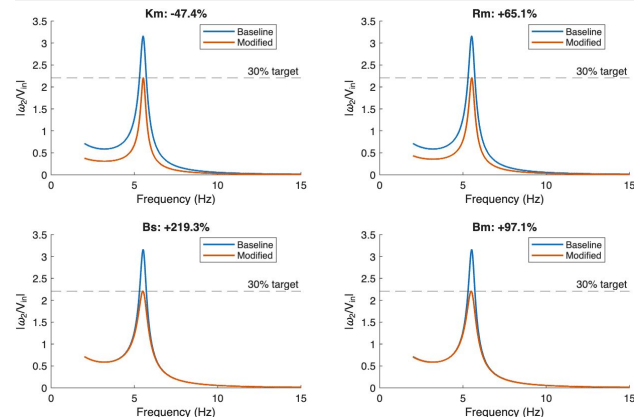
Metric	Experimental	Analytical / Model	Percent Difference
Resonant frequency	5.40 Hz	5.52 Hz	2.17%
Resonant amplitude	10.24 dB	9.62 dB	6.44%

Wheel Speed Analysis



Motor start-up excites the system's torsional resonance, causing the two flywheels to briefly rotate at different speeds. The flexible shaft transfers energy between them, producing a decaying oscillation until both reach the same steady-state speed. When the motor shuts off, the sudden loss of torque excites the torsional mode again, producing a second decaying transient.

Vibration Reduction



Increasing shaft stiffness improves the turn-on/off transient response, but it primarily shifts the torsional resonance rather than substantially reducing the resonant transmissibility peak

Design Recommendations

1. **Add motor-side damping:** increase B_m by 97% using a more damped motor-side coupling or torsional damper.
2. **Increase shaft-path damping:** increase B_s by 219% using a more dissipative shaft/coupling interface; effective, but requires a very large parameter change.
3. **Increase effective motor resistance:** increase R_m from about 2.06 to 3.40 Ω (+65%); easier to implement electrically, but would increase electrical losses and heat generation.
4. **Select a lower K_m motor:** reduce motor constant from about 0.112 to 0.059 V/(rad/s) (-47%); meets the target, but would likely require replacing or redesigning the motor.

Increasing shaft stiffness (K) by 118.6% reduced the peak turn-on/off relative-speed transient by 30%, but did not reduce the resonant transmissibility peak by 30%. This could be achieved by increasing shaft diameter, decreasing shaft length, or using a material with a higher shear modulus.

References

- Dorf, Richard C., and Robert H. Bishop. Modern Control Systems. 13th ed., Pearson, 2017.
- Toyota Technical Review: Contrasts in Value and Manufacturing—The Second-Generation Mirai and New GR Yaris. Vol. 66, no. 236, Toyota Motor Corporation, 22 Mar. 2021.
- Meriam, J. L., & Kraige, L. G. Engineering mechanics: Dynamics (6th ed.). Wiley. 2006.

- Experimental data acquired to determine resonant frequency and validate model
-

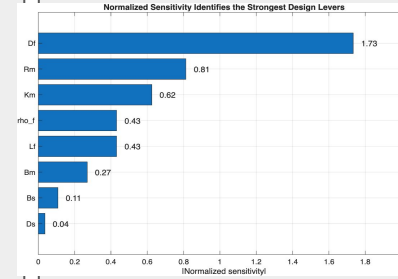
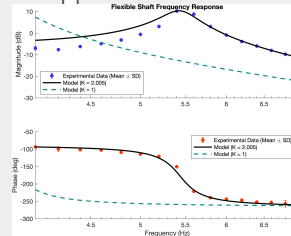
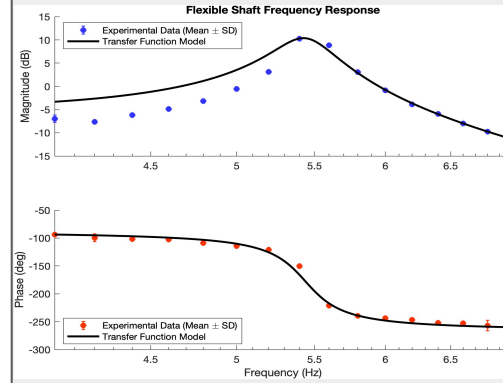
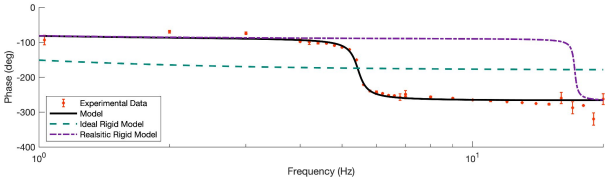


Figure 3. Normalized sensitivity analysis plot to determine which calculated parameters caused the most change in system response. The 4 largest normalized sensitivities were flywheel diameter (D_f), motor resistance (R_m), motor constant (K_m), and shaft density (ρ_{sh}) with values 1.73, 0.81, 0.62, and 0.43, respectively.

Effective communication:

- 1. adapt to your audience
- 2. maximize the signal-to-noise ratio
- 3. use effective redundancy



Our model matches up nicely with the experimental data on the bode plot. A slight deviation from the experimental data is present before the resonant frequency is reached because our model doesn't take into account non-viscous damping, which

Derived parameters

Parameter (units)	B_v	B_{θ}	B_s	B_b	B_m	D_s	L_s
Value	1.43E-03	4.20E-03	7.17E-04	7.17E-04	0	3.18E-03	3.05E-01

Parameter (units)	J_m	J_c	J_l	K	K	K_m	K_{theory}
Value	4.23E-05	3.45E-03	3.41E-03	2.06	2.01	1.12E-01	2.47

Peer Review Feedback

Vrinda: too much text - try to use short sentences/bullets and only call out key things - make images bigger/more detailed

System characterization and mathematical model

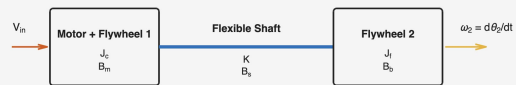
$$\frac{8.017 \cdot 10^{-5} s + 0.224}{2.418 \cdot 10^{-5} s^3 + 8.224 \cdot 10^{-5} s^2 + 0.0283 s + 0.0280}$$

Poster Checklist

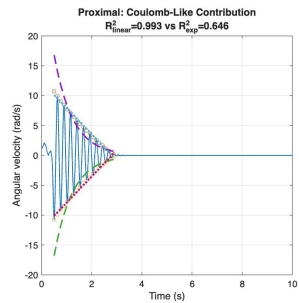
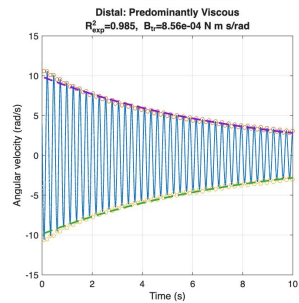
1. Visual Message
2. Title makes a claim and/or tells us the result
3. Text = 30/40% (text/labels are large enough, use margins & white space)
4. Supporting visuals = 60/70%
5. Limited branding
6. Descriptive headings, logical flow and order (guide the eye)
7. Avoid non-data ink
8. Visuals: resolution & clarity
9. Design so good you don't know it's there

Lec 2 notes

Flexible-Shaft Lumped-Parameter Model



Free-Response Tests Reveal Different Damping Behavior



Measured Frequency Response Reveals a 5.4 Hz Resonance

